Derivation of the scalar commutator bound

GPT Astra

The purpose of this note is to derive the bound

$$\boxed{\mathcal{D}_\phi \le 4r} \tag{1}$$

stated in the BMN section of *BPS Bootstrap*. We use the draft's creation-operator convention and normalization throughout. The proof consists of rewriting the BPS equation as a sum of squared norms. It gives the stronger intermediate result $3\mathcal{D}_\phi \le n_\phi + 2n_\psi$. The main argument is in Sections 1–3; Appendix A gives the details of reordering the operators.

1 Conventions

The unbarred matrices ϕ^i, ψ_i, f are creation operators, and bars denote their Hermitian conjugates. The flavor indices run from 1 to 3, with $\epsilon_{123} = 1$. Each matrix is divided by $\sqrt{N}$ relative to canonical oscillators, and $\mathrm{tr}(\cdot) = N^{-1}\,\mathrm{Tr}(\cdot)$. The matrices are traceless because the gauge group is $SU(N)$. We work in the large-N limit of the draft; equalities between moments below are understood in that limit.

The occupation numbers and charge are

$$n_\phi = \sum_i \left\langle \mathrm{tr}\Big(\phi^i \bar\phi^i\Big)\right\rangle, \quad n_\psi = \sum_i \left\langle \mathrm{tr}\Big(\psi_i \bar\psi_i\Big)\right\rangle, \quad n_f = \left\langle \mathrm{tr}\Big(f \bar f\Big)\right\rangle, \quad r = \frac{n_\phi}{3} + \frac{n_\psi}{6}. \tag{2}$$

For the $SU(3)_R$-invariant states considered in the draft, the three scalar occupations are equal, and so are the three fermion occupations. Thus these definitions agree with $n_\phi = 3\left\langle \mathrm{tr}\Big(\phi^1\bar\phi^1\Big)\right\rangle$ and $n_\psi = 3\left\langle \mathrm{tr}\Big(\psi_1\bar\psi_1\Big)\right\rangle$.

For any matrix polynomial $\mathcal{O}$, write

$$\|\mathcal{O}\|^2 = \left\langle \mathrm{tr}\Big(\mathcal{O}^\dagger \mathcal{O}\Big)\right\rangle \ge 0. \tag{3}$$

In particular,

$$3\mathcal{D}_\phi = \sum_{i<j} \left\langle \mathrm{tr}\Big([\phi^i,\phi^j][\bar\phi^j,\bar\phi^i]\Big)\right\rangle = \sum_{i<j} \left\| [\bar\phi^i,\bar\phi^j]\right\|^2. \tag{4}$$

The order in this formula matters: in general $\|\mathcal{O}\|^2 \ne \left\|\mathcal{O}^\dagger\right\|^2$.

2 A BPS identity written as squared norms

The supercharge in the draft is

$$\mathcal{Q} = -\sum_{i,j,k} \epsilon_{ijk}\,\mathrm{tr}\Big([\phi^j,\phi^k]\bar\psi_i\Big) + \sqrt{2}\sum_i \mathrm{tr}\Big([\phi^i,\psi_i]\bar f\Big). \tag{5}$$

Define the variation

$$\delta\mathcal{O} = \begin{cases} N^2[\mathcal{Q},\mathcal{O}], & \mathcal{O} \text{ bosonic,} \\ N^2\{\mathcal{Q},\mathcal{O}\}, & \mathcal{O} \text{ fermionic.} \end{cases} \tag{6}$$

The elementary commutation relations in Appendix A give

$$\delta\phi^i = 0, \qquad \delta\psi_i = -\sum_{j,k}\epsilon_{ijk}[\phi^j,\phi^k], \qquad \delta f = \sqrt{2}\sum_i[\phi^i,\psi_i], \tag{7}$$

$$\delta\bar\phi^i = 2\sum_{j,k}\epsilon_{ijk}[\phi^j,\bar\psi_k] - \sqrt{2}[\psi_i,\bar f], \qquad \delta\bar\psi_i = \sqrt{2}[\bar f,\phi^i], \qquad \delta\bar f = 0. \tag{8}$$

Taking the adjoint of (5) gives

$$\mathcal{Q}^\dagger = \sum_{i,j,k}\epsilon_{ijk}\,\mathrm{tr}\Big(\psi_i[\bar\phi^j,\bar\phi^k]\Big) + \sqrt{2}\sum_i \mathrm{tr}\Big(f[\bar\psi_i,\bar\phi^i]\Big). \tag{9}$$

A BPS state is annihilated by both $\mathcal{Q}$ and $\mathcal{Q}^\dagger$. Consequently, for any pure BPS state or any density matrix supported on BPS states,

$$0 = \Big\langle \delta\mathcal{Q}^\dagger \Big\rangle = N^2 \Big\langle \{\mathcal{Q},\mathcal{Q}^\dagger\} \Big\rangle. \tag{10}$$

We now evaluate $\Big\langle \delta\mathcal{Q}^\dagger \Big\rangle$ by varying the three letters in each term of (9). The contribution from varying ψ_i is

$$-\sum_{i,j,k,l,m}\epsilon_{ijk}\epsilon_{ilm}\Big\langle \mathrm{tr}\Big([\phi^l,\phi^m][\bar\phi^j,\bar\phi^k]\Big)\Big\rangle = -2\sum_{j,k}\Big\langle \mathrm{tr}\Big([\phi^j,\phi^k][\bar\phi^j,\bar\phi^k]\Big)\Big\rangle = 12\mathcal{D}_\phi. \tag{11}$$

Here we used $\sum_i \epsilon_{ijk}\epsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$, and the sum over j,k counts each unordered pair twice. The complete variation is therefore

$$\begin{aligned}\Big\langle \delta\mathcal{Q}^\dagger \Big\rangle &= 12\mathcal{D}_\phi - 2\sum_{i,j,k}\epsilon_{ijk}\Big\langle \mathrm{tr}\Big(\psi_i[\delta\bar\phi^j,\bar\phi^k]\Big)\Big\rangle \\ &\quad + 2\Big\|\sum_i[\bar\phi^i,\bar\psi_i]\Big\|^2 + 2\sum_i\Big\langle \mathrm{tr}\Big(f[[\bar f,\phi^i],\bar\phi^i]\Big)\Big\rangle - \sqrt{2}\sum_i\Big\langle \mathrm{tr}\Big(f\{\bar\psi_i,\delta\bar\phi^i\}\Big)\Big\rangle. \end{aligned}\tag{12}$$

The three terms on the second line come from varying f, $\bar\psi_i$, and $\bar\phi^i$, respectively, in the second term of $\mathcal{Q}^\dagger$. In particular, the variation of $[\bar\psi_i,\bar\phi^i]$ contains $-\{\bar\psi_i,\delta\bar\phi^i\}$, an anticommutator of the two fermionic matrices $\bar\psi_i$ and $\delta\bar\phi^i$.

Reordering the operators in (12) using the elementary commutation relations gives

$$-2\sum_{i,j,k}\epsilon_{ijk}\Big\langle \mathrm{tr}\Big(\psi_i[\delta\bar\phi^j,\bar\phi^k]\Big)\Big\rangle = 2\sum_{i,j,k}\epsilon_{ijk}\Big\langle \mathrm{tr}\Big([\psi_k,\bar\phi^j]\delta\bar\phi^i\Big)\Big\rangle - 8n_\psi, \tag{13}$$

$$-\sqrt{2}\sum_i\Big\langle \mathrm{tr}\Big(f\{\bar\psi_i,\delta\bar\phi^i\}\Big)\Big\rangle = -\sqrt{2}\sum_i\Big\langle \mathrm{tr}\Big([f,\bar\psi_i]\delta\bar\phi^i\Big)\Big\rangle - 6n_f, \tag{14}$$

$$\Big\langle \mathrm{tr}\Big(f[[\bar f,\phi^i],\bar\phi^i]\Big)\Big\rangle = \Big\|[\bar\phi^i,f]\Big\|^2 + n_f - 2\Big\langle \mathrm{tr}\Big(\phi^i\bar\phi^i\Big)\Big\rangle. \tag{15}$$

There is no sum over i in (15). Appendix A derives each identity directly from the elementary commutation relations.

The two quartic terms on the right of (13) and (14) combine into a squared norm, since

$$(\delta\bar{\phi}^i)^\dagger = 2\sum_{j,k}\epsilon_{ijk}[\psi_k,\bar{\phi}^j] - \sqrt{2}[f,\bar{\psi}_i]. \tag{16}$$

Substituting all three identities into (12) thus gives

$$\left\langle \delta\mathcal{Q}^\dagger \right\rangle = 12\mathcal{D}_\phi + \sum_i \left\|\delta\bar{\phi}^i\right\|^2 + 2\sum_i \left\|[\bar{\phi}^i,f]\right\|^2 + 2\left\|\sum_i[\bar{\phi}^i,\bar{\psi}_i]\right\|^2 \underbrace{-8n_\psi - 6n_f + 6n_f - 4n_\phi}_{-8n_\psi-4n_\phi}. \tag{17}$$

The terms proportional to n_f cancel. Dividing by 4 and using the BPS equation (10), we obtain

$$\boxed{3\mathcal{D}_\phi + \frac{1}{4}\sum_i \left\|\delta\bar{\phi}^i\right\|^2 + \frac{1}{2}\sum_i \left\|[\bar{\phi}^i,f]\right\|^2 + \frac{1}{2}\left\|\sum_i[\bar{\phi}^i,\bar{\psi}_i]\right\|^2 = n_\phi + 2n_\psi.} \tag{18}$$

3 The bound

Every squared norm in (18) is nonnegative. Dropping these terms proves

$$3\mathcal{D}_\phi \le n_\phi + 2n_\psi. \tag{19}$$

The last step uses only $n_\phi \ge 0$ and the definition of r:

$$\mathcal{D}_\phi \le \frac{n_\phi + 2n_\psi}{3} \le \frac{4n_\phi + 2n_\psi}{3} = 4r. \tag{20}$$

We can also write $4r - \mathcal{D}_\phi$ as a sum of nonnegative terms. The $SU(3)_R$ invariance of the state implies

$$\sum_i \left\|\delta\bar{\phi}^i\right\|^2 = 3\left\|\delta\bar{\phi}^3\right\|^2, \qquad \sum_i \left\|[\bar{\phi}^i,f]\right\|^2 = 3\left\|[\bar{\phi}^3,f]\right\|^2. \tag{21}$$

Substituting these relations into (18), using (2), and setting $i = 3$ in (8) gives

$$\boxed{4r - \mathcal{D}_\phi = n_\phi + \left\|[\phi^1,\bar{\psi}_2] - [\phi^2,\bar{\psi}_1] - \frac{[\psi_3,\bar{f}]}{\sqrt{2}}\right\|^2 + \frac{1}{2}\left\|[\bar{\phi}^3,f]\right\|^2 + \frac{1}{6}\left\|\sum_i[\bar{\phi}^i,\bar{\psi}_i]\right\|^2 \ge 0.} \tag{22}$$

In fact, with little extra effort, one can strengthen the bound to

$$\boxed{\mathcal{D}_\phi \le \min\left\{4r,\ r+\frac{3}{2},\ \frac{4j}{3},\ j+\frac{1}{2}\right\}.} \tag{23}$$

Here $j = n_\psi/2 + n_f$, as in the draft. The only additional reordering identity needed is

$$\sum_i \left\|[\bar{\phi}^i,f]\right\|^2 = \sum_i \left\|[\bar{f},\phi^i]\right\|^2 + 2n_\phi - 6n_f. \tag{24}$$

We derive it at the end of Appendix A. Substituting it into (18) cancels n_ϕ on the two sides and gives

$$3\mathcal{D}_\phi + \frac{1}{4}\sum_i \left\|\delta\bar{\phi}^i\right\|^2 + \frac{1}{2}\sum_i \left\|[\bar{f},\phi^i]\right\|^2 + \frac{1}{2}\left\|\sum_i[\bar{\phi}^i,\bar{\psi}_i]\right\|^2 = 3n_f + 2n_\psi. \tag{25}$$

Dropping the nonnegative norms in this identity and in (18) yields

$$3\mathcal{D}_\phi \le 2n_\psi + \min\{n_\phi, 3n_f\}. \tag{26}$$

To express this in terms of the charges, positivity of n_ϕ and n_f implies $n_\psi \le 6r$ and $n_\psi \le 2j$. The fermionic anticommutation relations also give

$$0 \le \sum_i \|\psi_i\|^2 = \sum_i \Big\langle \mathrm{tr}\Big(\bar\psi_i \psi_i\Big)\Big\rangle = 3 - n_\psi, \tag{27}$$

so $n_\psi \le 3$. Applying these restrictions to the two entries of (26) gives

$$\mathcal{D}_\phi \le \frac{n_\phi + 2n_\psi}{3} = r + \frac{n_\psi}{2} \le \min\left\{4r, r + \frac{3}{2}\right\}, \tag{28}$$

$$\mathcal{D}_\phi \le n_f + \frac{2n_\psi}{3} = j + \frac{n_\psi}{6} \le \min\left\{\frac{4j}{3}, j + \frac{1}{2}\right\}. \tag{29}$$

Together these prove (23). In particular, the upper bound is independent of r at fixed j once r is sufficiently large, and $\mathcal{D}_\phi = 0$ when $j = 0$.

A Details of the operator reordering

Latin indices $i, j, k, \ldots$ label flavors; Greek indices $\alpha, \beta, \gamma, \delta$ label colors. The nonzero elementary commutation relations are

$$\begin{aligned}
[(\bar\phi^i)^\alpha{}_\beta, (\phi^j)^\gamma{}_\delta] &= \frac{\delta^{ij}}{N}\left(\delta^\alpha{}_\delta \delta^\gamma{}_\beta - \frac{1}{N}\delta^\alpha{}_\beta \delta^\gamma{}_\delta\right), \\
[(\bar f)^\alpha{}_\beta, f^\gamma{}_\delta] &= \frac{1}{N}\left(\delta^\alpha{}_\delta \delta^\gamma{}_\beta - \frac{1}{N}\delta^\alpha{}_\beta \delta^\gamma{}_\delta\right), \\
\{(\bar\psi_i)^\alpha{}_\beta, (\psi_j)^\gamma{}_\delta\} &= \frac{\delta_{ij}}{N}\left(\delta^\alpha{}_\delta \delta^\gamma{}_\beta - \frac{1}{N}\delta^\alpha{}_\beta \delta^\gamma{}_\delta\right).
\end{aligned} \tag{A.1}$$

A trace over color indices is not cyclic when the matrix entries are noncommuting operators. For example, $\mathrm{tr}\Big(\bar\phi^i \phi^i\Big) - \mathrm{tr}\Big(\phi^i \bar\phi^i\Big) = 1 - 1/N^2$. We use (A.1) to evaluate the differences produced by reordering the operators in each trace.

A.1 The scalar–fermion term

First move $\bar\phi^k$ from the end of the trace to the beginning. For fixed i, j, k, the difference is

$$\begin{aligned}
\Big\langle \mathrm{tr}\Big(\psi_i\, \delta\bar\phi^j\, \bar\phi^k\Big) - \mathrm{tr}\Big(\bar\phi^k \psi_i\, \delta\bar\phi^j\Big)\Big\rangle &= \frac{1}{N}\sum_{\alpha,\beta,\gamma} \Big\langle (\psi_i)^\alpha{}_\beta [(\delta\bar\phi^j)^\beta{}_\gamma, (\bar\phi^k)^\gamma{}_\alpha]\Big\rangle \\
&= \frac{2}{N^2}\sum_{m,\alpha,\beta,\gamma} \epsilon_{jkm} \Big\langle (\psi_i)^\alpha{}_\beta \big((\bar\psi_m)^\beta{}_\alpha - \delta^\beta{}_\alpha (\bar\psi_m)^\gamma{}_\gamma\big)\Big\rangle \\
&= 2\sum_m \epsilon_{jkm}\left(\Big\langle \mathrm{tr}\Big(\psi_i \bar\psi_m\Big)\Big\rangle - \Big\langle \mathrm{tr}(\psi_i)\, \mathrm{tr}\Big(\bar\psi_m\Big)\Big\rangle\right).
\end{aligned} \tag{A.2}$$

Only the ϕ in the first term of $\delta\bar\phi^j$ can contract with $\bar\phi^k$. The two products in its matrix commutator give the two terms inside the parentheses. The double trace vanishes because the individual matrices are traceless.

After this move, expanding the commutator gives

$$\Big\langle \mathrm{tr}\Big(\psi_i[\delta\bar\phi^j,\bar\phi^k]\Big)\Big\rangle = -\Big\langle \mathrm{tr}\Big([\psi_i,\bar\phi^k]\delta\bar\phi^j\Big)\Big\rangle + 2\sum_m \epsilon_{jkm}\Big\langle \mathrm{tr}\Big(\psi_i\bar\psi_m\Big)\Big\rangle . \tag{A.3}$$

Multiply by $-2\epsilon_{ijk}$ and sum. The quadratic term becomes

$$-4\sum_{i,j,k,m}\epsilon_{ijk}\epsilon_{jkm}\Big\langle \mathrm{tr}\Big(\psi_i\bar\psi_m\Big)\Big\rangle = -8\sum_i\Big\langle \mathrm{tr}\Big(\psi_i\bar\psi_i\Big)\Big\rangle = -8n_\psi. \tag{A.4}$$

We used $\sum_{j,k}\epsilon_{ijk}\epsilon_{jkm} = 2\delta_{im}$. Relabeling the three dummy indices in the remaining term proves (13).

A.2 The fermion–field-strength term

Moving the last $\bar\psi_i$ to the front and using the fermionic anticommutation relations gives, for each fixed i,

$$\begin{aligned}\Big\langle \mathrm{tr}\Big(f\,\delta\bar\phi^i\,\bar\psi_i\Big) + \mathrm{tr}\Big(\bar\psi_i f\,\delta\bar\phi^i\Big)\Big\rangle &= \frac{1}{N}\sum_{\alpha,\beta,\gamma}\Big\langle f^\alpha{}_\beta\{(\delta\bar\phi^i)^\beta{}_\gamma,(\bar\psi_i)^\gamma{}_\alpha\}\Big\rangle\\ &= \frac{\sqrt{2}}{N^2}\sum_{\alpha,\beta,\gamma}\Big\langle f^\alpha{}_\beta((\bar f)^\beta{}_\alpha - \delta^\beta{}_\alpha(\bar f)^\gamma{}_\gamma)\Big\rangle\\ &= \sqrt{2}\Big(\Big\langle \mathrm{tr}\Big(f\bar f\Big)\Big\rangle - \Big\langle \mathrm{tr}(f)\,\mathrm{tr}\Big(\bar f\Big)\Big\rangle\Big) = \sqrt{2}n_f.\end{aligned} \tag{A.5}$$

Only $-\sqrt{2}[\psi_i,\bar f]$ in $\delta\bar\phi^i$ contributes to the anticommutator. It follows that

$$\Big\langle \mathrm{tr}\Big(f\{\bar\psi_i,\delta\bar\phi^i\}\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big([f,\bar\psi_i]\delta\bar\phi^i\Big)\Big\rangle + \sqrt{2}n_f. \tag{A.6}$$

Multiplying by $-\sqrt{2}$ and summing the three flavors gives (14), with the quadratic term $-6n_f$.

A.3 The scalar–field-strength term

Fix a flavor i; no sum over i is understood until the final step. Expanding the double commutator gives

$$\Big\langle \mathrm{tr}\Big(f[[\bar f,\phi^i],\bar\phi^i]\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big(f\bar f\phi^i\bar\phi^i\Big)\Big\rangle - \Big\langle \mathrm{tr}\Big(f\phi^i\bar f\bar\phi^i\Big)\Big\rangle - \Big\langle \mathrm{tr}\Big(f\bar\phi^i\bar f\phi^i\Big)\Big\rangle + \Big\langle \mathrm{tr}\Big(f\bar\phi^i\phi^i\bar f\Big)\Big\rangle . \tag{A.7}$$

The squared norm to which we compare it is

$$\Big\|[\bar\phi^i,f]\Big\|^2 = \Big\langle \mathrm{tr}\Big(\bar f\phi^i\bar\phi^i f\Big)\Big\rangle - \Big\langle \mathrm{tr}\Big(\bar f\phi^i f\bar\phi^i\Big)\Big\rangle - \Big\langle \mathrm{tr}\Big(\phi^i\bar f\bar\phi^i f\Big)\Big\rangle + \Big\langle \mathrm{tr}\Big(\phi^i\bar f f\bar\phi^i\Big)\Big\rangle . \tag{A.8}$$

The four elementary reordering relations needed to compare the two lines are

$$\Big\langle \mathrm{tr}\Big(\bar f\phi^i\bar\phi^i f\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big(f\bar f\phi^i\bar\phi^i\Big)\Big\rangle + \Big\langle \mathrm{tr}\Big(\phi^i\bar\phi^i\Big)\Big\rangle , \tag{A.9}$$

$$\Big\langle \mathrm{tr}\Big(\bar f\phi^i f\bar\phi^i\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big(f\bar\phi^i\bar f\phi^i\Big)\Big\rangle , \tag{A.10}$$

$$\Big\langle \mathrm{tr}\Big(\phi^i\bar f\bar\phi^i f\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big(f\phi^i\bar f\bar\phi^i\Big)\Big\rangle , \tag{A.11}$$

$$\Big\langle \mathrm{tr}\Big(\phi^i\bar f f\bar\phi^i\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big(f\bar\phi^i\phi^i\bar f\Big)\Big\rangle + \Big\langle \mathrm{tr}\Big(\phi^i\bar\phi^i\Big)\Big\rangle - n_f. \tag{A.12}$$

As throughout the note, these relations are at leading large N. In (A.9), moving the last f to the front contracts it with $\bar{f}$, leaving $\Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle$. In (A.10), move the initial pair $\bar{f}\phi^i$ to the end; the terms generated by the commutation relations are products of one-letter traces, which vanish. In (A.11), moving the last f to the front likewise leaves only such a product. In (A.12), first move ϕ^i to the end and then move $\bar{f}$ to the end. These two steps produce, respectively, $-\Big\langle \mathrm{tr}\Big(\bar{f}f\Big)\Big\rangle$ and $+\Big\langle \mathrm{tr}\Big(\bar{\phi}^i\phi^i\Big)\Big\rangle$. Their sum is

$$-\Big\langle \mathrm{tr}\Big(\bar{f}f\Big)\Big\rangle + \Big\langle \mathrm{tr}\Big(\bar{\phi}^i\phi^i\Big)\Big\rangle = -(n_f+1) + (\Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle + 1) = \Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle - n_f. \tag{A.13}$$

The vacuum constants cancel.

Substitute (A.9)–(A.12) into (A.8). Its four quartic terms become precisely (A.7), while the quadratic terms add up to $2\Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle - n_f$. Therefore

$$\Big\|[\bar{\phi}^i, f]\Big\|^2 = \Big\langle \mathrm{tr}\Big(f[[\bar{f},\phi^i],\bar{\phi}^i]\Big)\Big\rangle + 2\Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle - n_f, \tag{A.14}$$

which is (15).

The same calculation also relates the two orderings of the squared norm. Moving the last $\bar{\phi}^i$ to the front gives

$$\Big\langle \mathrm{tr}\Big(f[\bar{f},\phi^i]\bar{\phi}^i\Big) - \mathrm{tr}\Big(\bar{\phi}^i f[\bar{f},\phi^i]\Big)\Big\rangle = -n_f. \tag{A.15}$$

Only the ϕ^i contracts with $\bar{\phi}^i$, giving $-\Big\langle \mathrm{tr}\Big(f\bar{f}\Big)\Big\rangle$; the remaining product of one-letter traces vanishes. Consequently,

$$\Big\langle \mathrm{tr}\Big(f[[\bar{f},\phi^i],\bar{\phi}^i]\Big)\Big\rangle = \Big\langle \mathrm{tr}\Big([\bar{\phi}^i, f][\bar{f},\phi^i]\Big)\Big\rangle - n_f = \Big\|[\bar{f},\phi^i]\Big\|^2 - n_f. \tag{A.16}$$

Comparing with (15) gives

$$\Big\|[\bar{\phi}^i, f]\Big\|^2 = \Big\|[\bar{f},\phi^i]\Big\|^2 + 2\Big\langle \mathrm{tr}\Big(\phi^i\bar{\phi}^i\Big)\Big\rangle - 2n_f. \tag{A.17}$$

Summing over the three flavors proves (24).